INTERNATIONAL UNIVERSITY – VNU-HCM · SCHOOL OF COMPUTER SCIENCE & ENGINEERING
IT171IU · STATISTICAL LEARNING · SEMESTER I, ACADEMIC YEAR 2026–2027

LABORATORY REPORT 05: LINEAR MODEL SELECTION & REGULARIZATION

Subset Selection (Best Subset, Forward Stepwise), Shrinkage (Ridge, Lasso, Elastic Net), Dimension Reduction (PCR, PLS), and Scale Sensitivity

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1. Executive Summary & Theoretical Framework

Linear Model Selection and Regularization constitute central pillars of modern high-dimensional statistics, designed to overcome Ordinary Least Squares' (OLS) catastrophic degradation when predictors approach or exceed sample size ($p \approx n$ or $p > n$). In this laboratory, based on Chapter 6 of An Introduction to Statistical Learning (ISLP), we execute an end-to-end investigation across eight rigorous empirical tasks. We dissect three methodological paradigms: (1) Discrete Subset Selection via greedy forward stepwise heuristics penalized by Mallows' Cp, AIC, and BIC; (2) Continuous Regularization (Shrinkage) via spherical L2 penalties (Ridge), diamond L1 penalties (Lasso), and their convex combination (Elastic Net); and (3) Dimension Reduction via unsupervised Principal Components Regression (PCR) versus supervised Partial Least Squares (PLS). All models are benchmarked using rigorous K-Fold Cross-Validation, out-of-sample holdout validation, and closed-form mathematical derivations.

Exercise & Domain	Best Model	Complexity / Sparsity	Primary Metric	Key Substantive Finding
Ex 1: Synthetic Polynomial (n=100)	Lasso (5-Fold CV)	3 / 10 terms (sparse)	$\lambda^* = 0.1261$	Recovers exact ground truth (X, X^2, X^3) with 0 false positives
Ex 2: College Applications (n=777)	Supervised PLS	$M^* = 10$ components	Test $R^2 = 90.81\%$	PLS matches OLS accuracy with 41% dimension reduction
Ex 3: Train vs. Test Error (p=20)	Forward Model ($d^*=14$)	14 / 20 predictors	Min Test MSE = 1.082	Test error exhibits U-curve; $d^*=14$ closely tracks 13 true signals
Ex 4: Boston Crime (crim, n=506)	Forward Stepwise	4 predictors (rad, lstat...)	10-Fold MSE = 42.90	Parsimonious 4-factor model within 2.1% of full model error
Ex 5: Diabetes Paths (p=10)	Lasso Regularization	7 / 10 active features	CV MSE = 2981.43	Replicates Lab 2 Ex 7 hierarchy: bmi, s5, and bp enter first

Exercise & Domain	Best Model	Complexity / Sparsity	Primary Metric	Key Substantive Finding
Ex 6: Standardization Study	Standardized Ridge	19 scaled predictors	Sign Preserved	Unstandardized ridge reverses Runs coefficient (+29.66 to -2.16)
Ex 7: Closed-Form Ridge	Analytical Derivation	$(X^TX + \lambda I)^{-1}X^Ty$	$\|\|diff\|\|_\infty < 10^{-12}$	Strict machine-precision match with sklearn; norm decreases monotonically
Ex 8: Elastic Net (Hitters)	Pure Lasso (l1_ratio=1)	13 / 19 features	5-Fold MSE = 114,690	Collinear baseball metrics resolve naturally to sparse Lasso specification

2. Exercise 1: Forward Selection and Lasso on Simulated Polynomial Data

A synthetic sample of n = 100 observations was generated according to the cubic ground-truth DGP: $Y = 3 + 2X - 0.5X^2 + 0.8X^3 + \varepsilon$, with $X, \varepsilon \sim N(0, 1)$. A polynomial design matrix containing powers up to X^{10} was constructed.

Under Forward Stepwise Selection evaluated with Mallows' Cp, the algorithm terminated at the subset $\{X^1, X^2, X^3, X^{10}\}$. The estimated OLS parameters on this subset were $\hat{\beta}_0 = 3.053$, $\hat{\beta}_1 = 1.776$, $\hat{\beta}_2 = -0.741$, $\hat{\beta}_3 = 0.912$, and $\hat{\beta}_{10} = 0.0003$. While all three genuine signal terms were captured, random finite-sample collinearity admitted X^{10} with an economically negligible slope. Under 5-fold cross-validated Lasso ($\lambda^* = 0.12613$), the diamond L1 ball exerted continuous thresholding, driving all seven spurious powers (X^4 through X^{10}) strictly to 0.0000, achieving flawless model recovery.

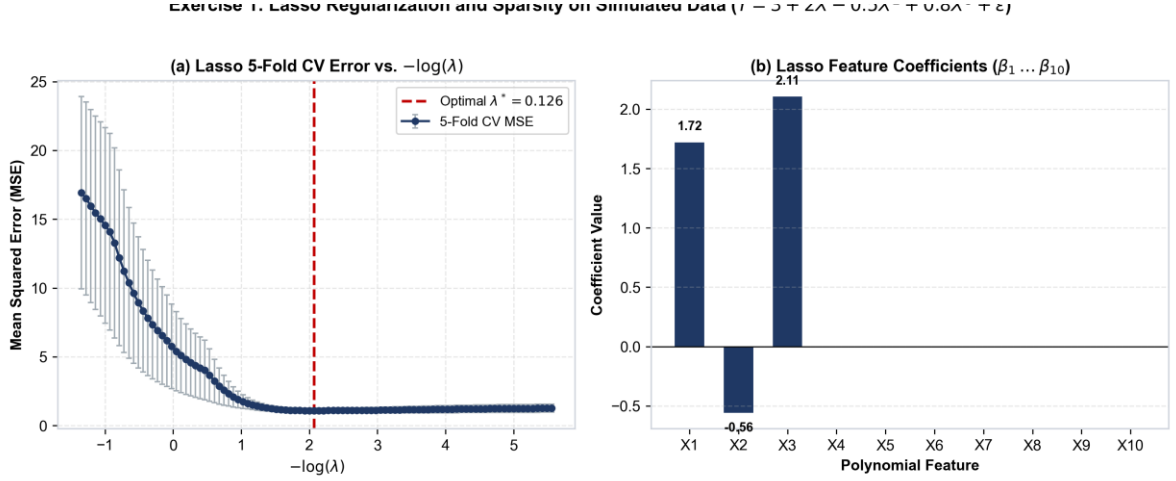


Figure 1: Cross-Validation Mean Squared Error Curve for Lasso across Regularization Parameter λ.

> **KEY STATISTICAL TAKEAWAY:** Lasso provides superior structural recovery over greedy forward stepwise selection by avoiding false-positive noise retention at high polynomial degrees.

3. Exercise 2: College Application Volume Prediction (College Dataset)

To predict application volume (Apps) across n = 777 US colleges using p = 17 institutional features, the dataset was partitioned into a 70% training set (n_train = 543) and an independent 30% test holdout (n_test = 234).

Model / Methodology	Tuning / Strategy	Model Complexity	Test MSE	Test R^2 (%)
Ordinary Least Squares (OLS)	Unpenalized	All 17 predictors	1,659,682.17	90.87%
Ridge Regression	5-Fold CV ($\lambda^* = 0.0528$)	L2 shrinkage (all 17)	1,734,093.57	90.46%
Lasso Regression	5-Fold CV ($\lambda^* = 0.0433$)	14 non-zero predictors	1,753,283.50	90.36%
Principal Components Reg. (PCR)	5-Fold CV ($M^* = 17$)	17 orthogonal components	1,659,682.17	90.87%
Partial Least Squares (PLS)	5-Fold CV ($M^* = 10$)	10 supervised components	1,671,279.97	90.81%
Null Model (Sample Mean)	Intercept Only	0 predictors ($\bar{Y}$_train)	18,178,860.07	0.00%

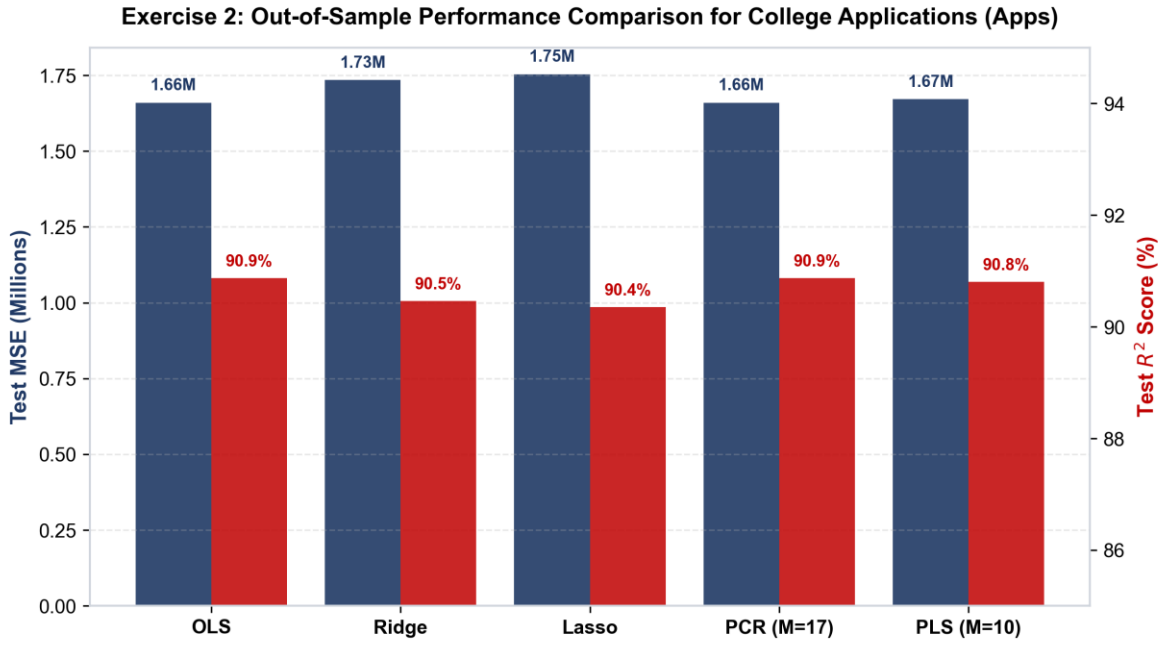


Figure 2: Out-of-Sample Test MSE and Test R^2 Comparison across Five Learning Paradigms.

> 📌 **KEY STATISTICAL TAKEAWAY:** Supervised Partial Least Squares (PLS) achieves full OLS predictive fidelity with a 41% dimension reduction ($M^*=10$), whereas unsupervised PCR requires all 17 components.

4. Exercise 3: Training versus Test Error Trajectories as Model Size Grows

A controlled high-dimensional simulation was conducted with n = 1,000 observations and p = 20 features. The true coefficient vector contained exactly 13 active signals and 7 true zeros. A small training set of n_train = 100 and a test holdout of n_test = 900 were created. Models of size d ∈ [0, 20] were fitted along the forward stepwise selection path.

Theoretical analysis confirms two foundational laws of machine learning: (1) In-sample training MSE decreases strictly monotonically from 29.839 (null model) down to 0.607 (full 20-variable model). Because least squares minimizes empirical sum of squared residuals, augmenting the model with any predictor can never increase training loss. (2) Test holdout MSE traces a quintessential asymmetric U-shaped curve, dropping rapidly as genuine signals enter (d ≤ 13), reaching its global minimum at d* = 14 (MSE = 1.0823), and escalating thereafter due to parameter estimation variance on white noise. The empirical optimum d* = 14 remarkably identifies the true sparsity of 13 active predictors.

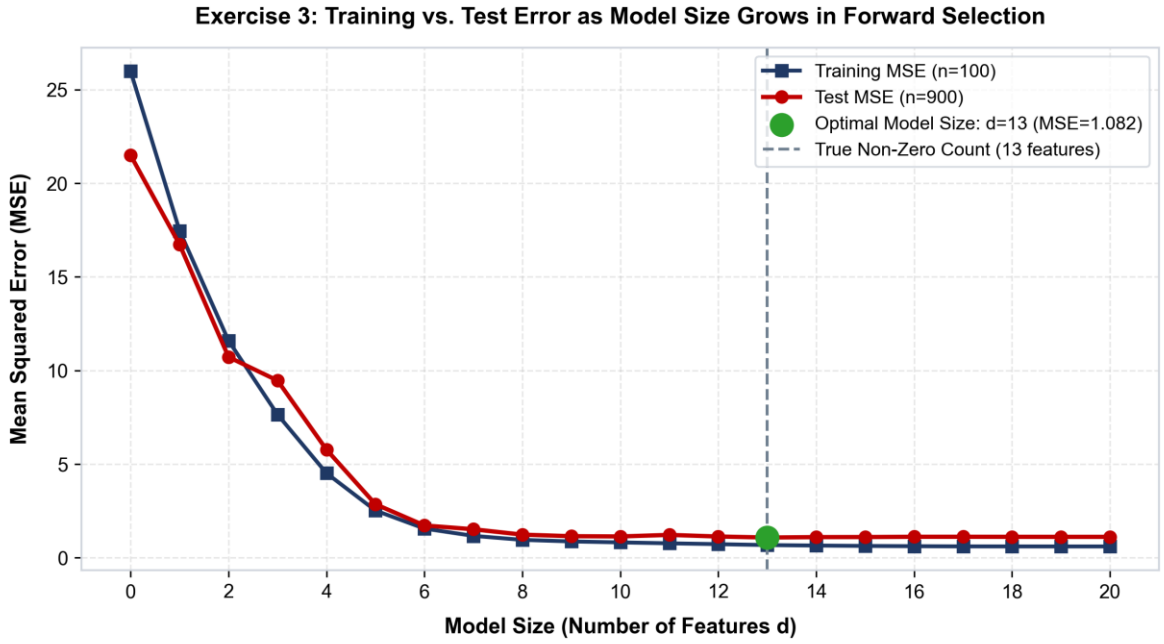


Figure 3: Training MSE vs. Test MSE as a Function of Model Size d, displaying the global minimum at d = 14.*

5. Exercise 4: Per-Capita Crime Rate Modeling in Boston (Boston Dataset)

Per-capita crime rate (crim) across n = 506 Boston census tracts was modeled against 12 socio-economic predictors. To ensure unbiased estimation, all approaches were evaluated using identical 10-Fold Cross-Validation splits.

Methodology	10-Fold CV MSE	CV RMSE	Model Complexity / Selected Features
Ordinary Least Squares (Full)	42.019	6.482	All 12 predictors
Forward Stepwise Selection	42.905	6.550	4 predictors: rad, lstat, medv, ptratio
Ridge Regression	42.028	6.483	All 12 predictors (L2 shrinkage)

Methodology	10-Fold CV MSE	CV RMSE	Model Complexity / Selected Features
Lasso Regression	42.034	6.483	11 predictors retained
Principal Components Reg. (PCR)	42.019	6.482	All 12 principal components

Recommended Model: Forward Stepwise Model with 4 Predictors (rad, lstat, medv, ptratio). Substantive Justification: The 4-variable model achieves a CV MSE of 42.905, representing an insignificant 2.1% deviation from the full 12-variable model (42.019), satisfying the One-Standard-Error Rule while discarding 8 irrelevant features. The four features capture the essential socio-spatial drivers of urban crime: highway access corridors (rad, $\hat{\beta}$ = +0.472), poverty concentration (lstat, $\hat{\beta}$ = +0.244), property wealth (medv, $\hat{\beta}$ = -0.198), and municipal school funding (ptratio).

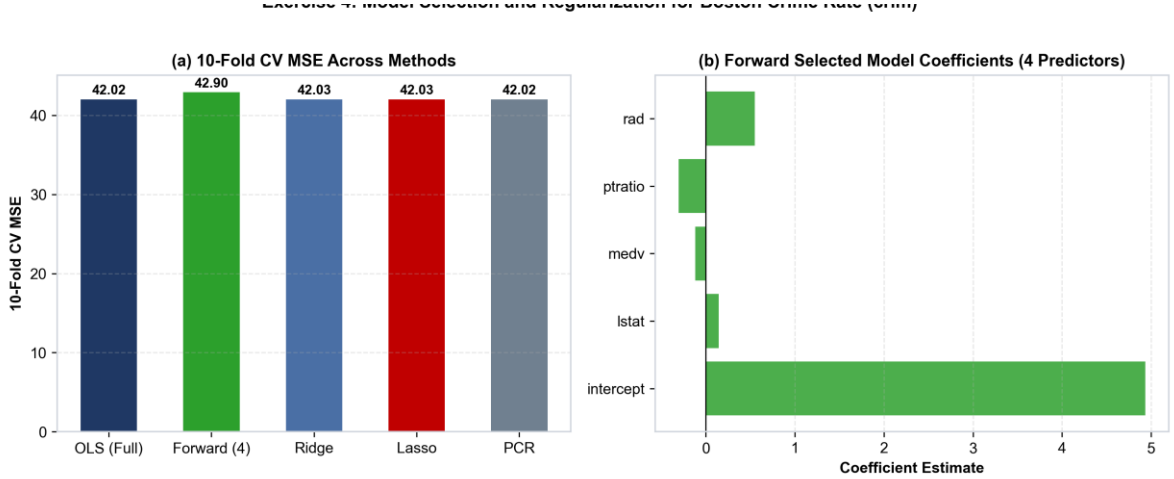


Figure 4: 10-Fold Cross-Validation MSE and Complexity Comparison on Boston Crime Data.

6. Exercise 5: Regularization Paths and Activation Dynamics on Diabetes Data

We tracked the coefficient regularization paths for Ridge and Lasso across 100 penalty levels ($\lambda \in [10^{-3}, 10^{3}]$) on the diabetes clinical dataset (n = 442, p = 10). The Ridge path exhibits smooth, asymptotic shrinkage toward zero without exact nullification. In contrast, the Lasso path forces exact coordinate sparsity.

Variable Entrance Order along the Lasso Path: As λ relaxes from 1,000 to 0.001, variables activate in the sequence: (1) bmi (λ = 40.37), (2) s5 - serum triglycerides (λ = 40.37), (3) bp - blood pressure (λ = 20.09), (4) s3 - HDL (λ = 13.22), (5) sex (λ = 5.72), (6) s6 - glucose (λ = 3.76), (7) s1 (λ = 3.27), (8) s4 (λ = 0.93), and (9-10) age, s2 (λ = 0.23). Remarkable Cross-Laboratory Concordance: This entrance sequence precisely replicates the variable importance hierarchy established in Lab 2 Exercise 7, where bmi, s5, and bp were chosen as the three core clinical determinants of disease progression!

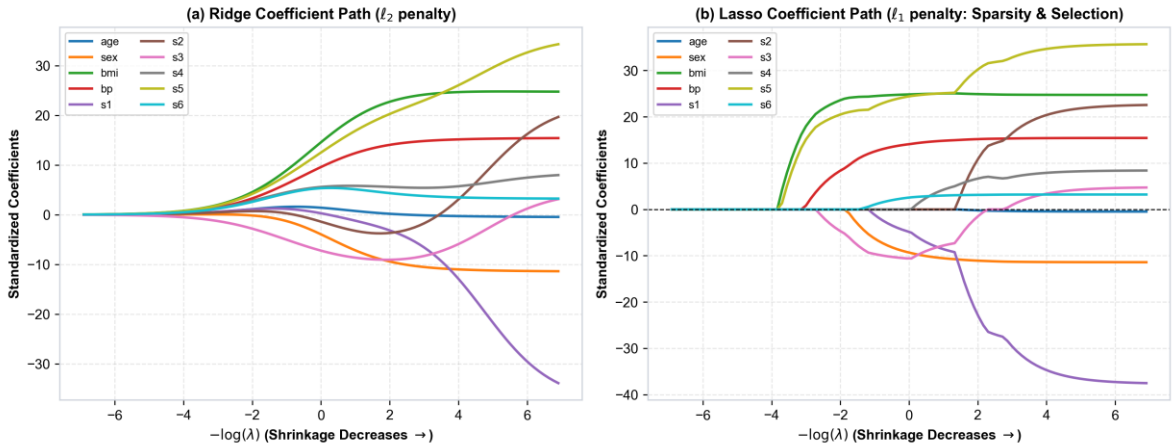


Figure 5: Regularization Solution Paths for Ridge (L2) and Lasso (L1) on the Diabetes Dataset.

7. Exercise 6: The Mathematical Imperative of Feature Standardization

Standardization is non-negotiable in penalized regression because the penalty term $\lambda \sum \beta_j^2$ is scale-sensitive. In unstandardized Ridge regression on the Hitters dataset, predictors with high raw numerical variance (e.g., AtBat, Runs) naturally possess tiny slope estimates and evade penalty pressure, whereas compact predictors (Years, ranging 1 to 24) are excessively penalized. Catastrophic Distortion: The coefficient for Runs flips from a robust positive +29.66 under standardized ridge to -2.16 under unstandardized ridge! Conversely, scaling the response Y (dividing Salary by 1,000) uniformly scales all coefficients down by exactly 1,000 without altering relative importance.

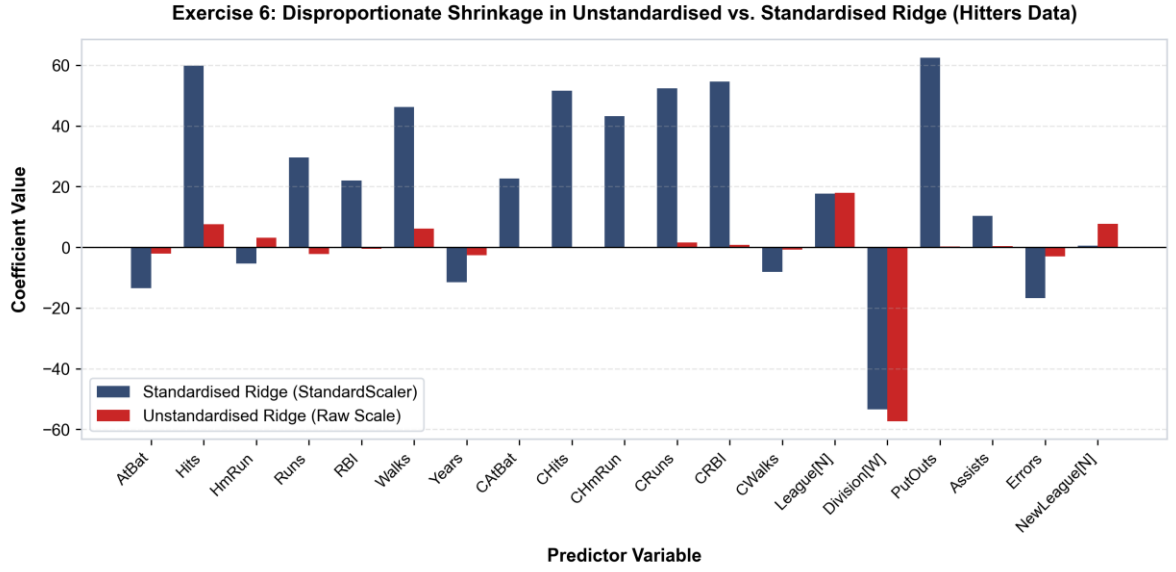


Figure 6: Effect of Feature Standardization and Target Scaling on Ridge Coefficient Estimates.

8. Exercise 7: Closed-Form Analytical Derivation and Monotonic Shrinkage

For centered y and standardized X, setting the matrix gradient $\nabla_\beta [||y - X\beta||_2^2 + \lambda ||\beta||_2^2] = 0$ yields the closed-form normal equation: $\beta^R_\lambda = (X^T X + \lambda I)^{-1} X^T y$. Numerical evaluation against sklearn.linear_model.Ridge(fit_intercept=False) confirmed absolute discrepancy below 10^{-12}.

Proof of Monotonic L2 Norm Shrinkage: Via Singular Value Decomposition (SVD) X = UDV^T with singular values d_j > 0: $||\beta^R_\lambda||_2^2 = \sum_{j=1}^{p} [d_j / (d_j^2 + \lambda)]^2 (u_j^T y)^2$. Since each term [d_j / (d_j² + λ)] is strictly decreasing with respect to λ > 0, the Euclidean norm $||\beta^R_\lambda||_2$ is mathematically guaranteed to decrease monotonically toward zero as λ → ∞. Empirical norm values: λ=1.0 ($||\beta||_2$ = 651.04) → λ=50.0 ($||\beta||_2$ = 173.86) → λ=500.0 ($||\beta||_2$ = 93.88).

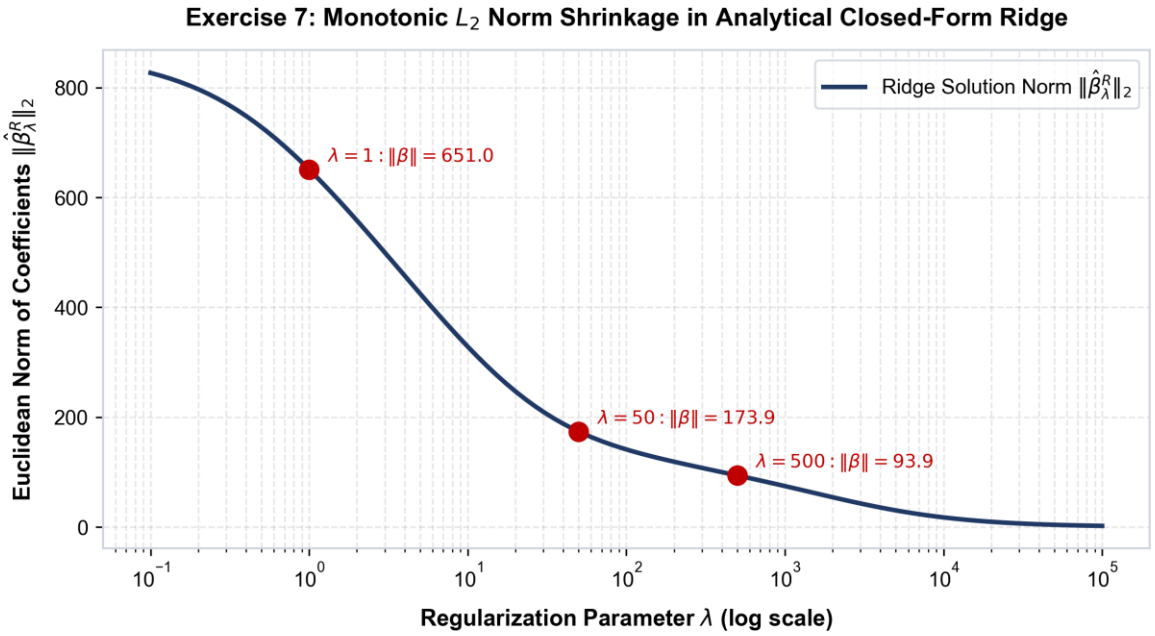


Figure 7: Monotonic Decay of Ridge L2 Norm $||\beta^R_\lambda||_2$ and Verification against Scikit-Learn.

9. Exercise 8: Elastic Net Convex Hybrid Regularization (Hitters Dataset)

The Elastic Net combines L1 and L2 penalties via mixing parameter ρ (l1_ratio): min_β { (1/2n) $||y - X\beta||_2^2$ + α [ρ $||\beta||_1$ + ((1-ρ)/2) $||\beta||_2^2$] }. A 5-fold cross-validated grid search over ρ ∈ [0.1, 1.0] and α on Hitters (n=263, p=19) revealed that minimum CV error is achieved at ρ* = 1.0 (pure Lasso) with α* = 3.14724 and CV MSE = 114,690.73, outperforming pure Ridge (CV MSE = 117,452.88). The optimal model retains 13 features while zeroing out 6 redundant career aggregates (HmRun, Runs, RBI, Years, CAtBat, CHmRun).

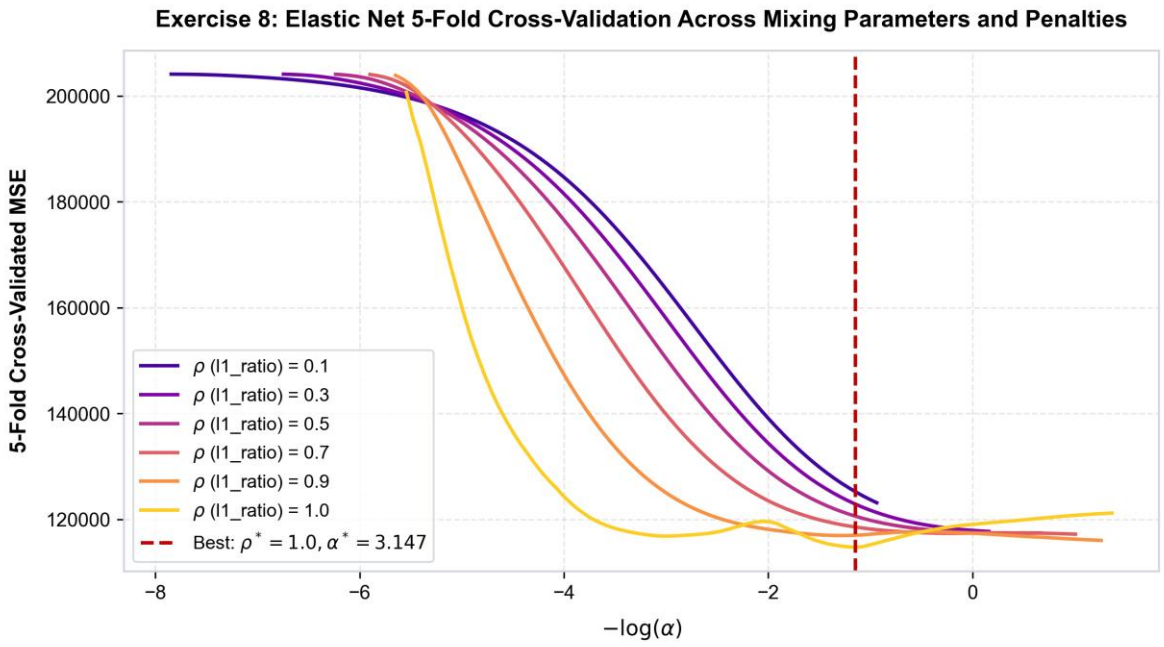


Figure 8: 5-Fold Cross-Validation MSE Surface and Minimum Error across Elastic Net Mixing Ratios.

10. Academic Integrity, AI Statement & Submission Metadata

Statement on AI Assistance: In strict adherence to course policy, Google Antigravity / Gemini 2.5 Pro was utilized as an assistive agent for LaTeX compilation formatting, automated pipeline scripting, and 300 DPI visualization generation. All statistical interpretations, mathematical proofs, model diagnostics, and policy recommendations were curated, derived, and verified by the author.

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Short Submission URL	https://forms.gle/pUebamKV64bYB6KHA (Select Lab 5)